

OPERATING SYSTEMS: COMPREHENSIVE MASTER TEXTBOOK

1. Introduction to OS

What is an Operating System?

An OS is an intermediary between a computer user and the computer hardware. Its primary goals are to execute user programs, make the computer system easy to use, and utilize hardware efficiently.

2. Process Management

Processes & Scheduling

A process is a program in execution. The OS manages processes via **CPU Scheduling** (e.g., FCFS, Round Robin, SJF). It ensures the CPU is utilized effectively while minimizing waiting time for user tasks.

3. Memory Management

Paging and Segmentation

The OS maps logical addresses (used by processes) to physical addresses (RAM). **Paging** breaks memory into fixed-size blocks (pages/frames), while **Segmentation** groups memory into logical units (code, data, stack).

4. Deadlocks

Handling Deadlocks

A deadlock occurs when a set of processes are blocked because each process is holding a resource and waiting for another resource held by another process. OS handles this via prevention, avoidance (Banker's Algorithm), or detection/recovery.

5. File Systems

Storage Organization

The file system provides the mechanism for on-disk storage and access to data. It manages directory structures, file allocation methods (contiguous, linked, indexed), and free-space management.